

Measuring error

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Warmup

Your friend collects $n = 100$ observations with a binary response, and fits a logistic regression model:

$$Y_i \sim \text{Bernoulli}(p_i)$$
$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{99} X_{i,99}$$

Using this data, they then create a confusion matrix:

Predicted Status	Actual Status	
	negative	positive
negative	45	0
positive	0	55

What has gone wrong?

$n=100$ $K+1 = 100$
(same # of observations as parameters)

Linear regression:

$$\hat{y} = X(X^T X)^{-1} X^T y \quad \in \text{colspace}(X)$$

Suppose $\text{ncol}(X) = n$ ($= \text{length}(y)$)

$$(X \in \mathbb{R}^{n \times n}) \quad (y \in \mathbb{R}^n)$$

Then: if columns of X are linearly independent,
 $\text{colspace}(X) = \mathbb{R}^n \Rightarrow y \in \text{colspace}(X)$

$$\dim(\text{colspace}(X)) = n$$

$$\Rightarrow \hat{y} = y$$

(perfect interpolation on
training data,

not useful for predicting
new data)

Take-aways from last class activity

- ▶ Even if the model is correctly specified, prediction performance on the **training** data is a biased estimate of prediction performance on **new** data
- ▶ The bias gets worse as the number of explanatory variables in the model increases!

Terminology and notation

Training data:

$$\mathcal{T} = \{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

$(x_i, y_i) \sim g$ (typically unknown)

$\mathcal{X} \subseteq \mathbb{R}^d$ support of x

$\mathcal{Y} \subseteq \mathbb{R}$ support of y

Fitted model:

$\mathcal{T}$ used to estimate some function h
regression: $\hat{h}(x)$ estimates $\mathbb{E}[y|x]$

classification: $\hat{h}(x) \in \mathcal{Y}$ has same support as $\mathcal{Y}$
e.g. binary response $\rightarrow \hat{h}(x) \in \{0, 1\}$

New data:

$$(x^*, y^*)$$

Loss functions

$$(x, y) \sim g$$

Loss function: $L(y, \hat{h}(x))$ measures how close $\hat{h}(x)$ is to y

Examples:

Squared error loss: $L(y, \hat{h}(x)) = (y - \hat{h}(x))^2$

Absolute error loss: $L(y, \hat{h}(x)) = |y - \hat{h}(x)|$

Log-likelihood loss: use negative log-likelihood
e.g. in binary data,

$$L(y, \hat{h}(x)) = -y \log(\hat{h}(x)) - (1-y) \log(1 - \hat{h}(x))$$

0-1 loss: (when y & $\hat{h}(x)$ are discrete)

$$L(y, \hat{h}(x)) = \mathbb{1}\{y \neq \hat{h}(x)\}$$

Risk (error)

Training error: $\mathcal{T} = \{(x_1, y_1), \dots, (x_n, y_n)\}$

$$R_{\text{train}}(\hat{h}) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{h}(x_i))$$

Empirical risk minimization: choose $\hat{h}$ to minimize $R_{\text{train}}(\hat{h})$

Test error: new sample $(x^*, y^*) \sim q$

$$R_{\mathcal{T}}(\hat{h}) = \mathbb{E}_{x^*, y^*} [L(y^*, \hat{h}(x^*)) \mid \mathcal{T}]$$

$\uparrow$
conditional on $\mathcal{T}$

$\uparrow$
 $\mathbb{E}$ is not
wrt $\hat{h}$

$\uparrow$ conditional on $\mathcal{T}$

Expected test error:

$$R = \mathbb{E}_{\mathcal{T}} [R_{\mathcal{T}}(\hat{h})]$$

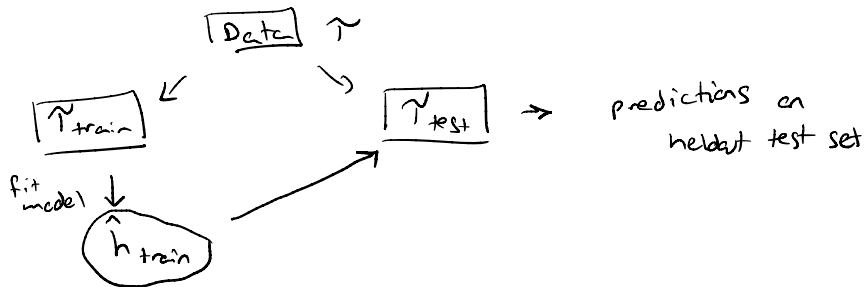
(expectation of test error over all possible training sets)

What do we do??

If training error is a **biased** estimate for error on new data, how do we proceed?

Data splitting

- ▶ Divide the available data $\mathcal{T}$ into two disjoint subsets: $\mathcal{T}_{\text{train}}$ and $\mathcal{T}_{\text{test}}$.
- ▶ Common train/test splits include 60/40, 70/30, 80/20



k-fold cross validation

- ▶ Divide the available data $\mathcal{T}$ into k disjoint subsets (called **folds**) $\mathcal{T}_1, \dots, \mathcal{T}_k$



Heldout : $\mathcal{T}_1$

Train $\mathcal{T}_2, \dots, \mathcal{T}_k \rightarrow \hat{h}_{-1}$

Heldout : $\mathcal{T}_2$

Train $\mathcal{T}_1, \mathcal{T}_3, \dots, \mathcal{T}_k \rightarrow \hat{h}_{-2}$

$$R_{cv} = \frac{1}{n} \sum_{i=1}^n L(\gamma_i, \hat{h}_{-j(i)}(x_i))$$

Leave-one-out (V):

$k=n$

$j(i)$ denote field
to which
observation i belongs

Class activity

Work on the class activity (course website)